

Akash Maity

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OBJECTIVE

Computer Science graduate (Dalhousie University, 2023–2026) specializing in machine learning, deep learning, and generative AI. Experienced across reinforcement learning, sequence modeling, retrieval-augmented generation, and autonomous AI agents, with applied work spanning research pipelines, explainable ML audits, and production-facing tools. Currently a research collaborator with the Ocean Tracking Network and Technical Project Lead at the Dalhousie Machine Learning Society. Seeking full-time Machine Learning Engineering or Data Science / ML roles with a **January 2027** start date.

EDUCATION

Bachelor of Computer Science

2023 – 2026

Dalhousie University, Halifax, NS

- Hackathons: **ShiftKey Labs Hackathon** — built PSI, a marine digital twin platform for offshore wind siting; **Watt the Hack** — built a sustainability and green-efficiency web app (React, TypeScript, Supabase)

TECHNICAL SKILLS

Languages & Frameworks

Python, SQL, JavaScript, TypeScript

Machine Learning & DL

Supervised/unsupervised learning; reinforcement learning (Q-learning, SARSA, DQN, CQL, PPO, Actor-Critic); transfer learning; RNN/GRU/LSTM sequence models; Kalman filtering; recommender systems; ANN semantic search

Generative AI & Agents

Autonomous AI agents; fine-tuning & customizing pre-trained LLMs; RAG (LangChain, FAISS); Hugging Face transformers; NLI-based hallucination detection; prompt engineering; SHAP, LIME, partial dependence plots

Data Engineering & Cloud

ETL design, pipeline orchestration, data preprocessing & feature engineering, automated evaluation/monitoring harnesses, synthetic data generation, AWS, Google Cloud Platform, Firebase, Cloudflare R2, Power BI, REST API integration

RESEARCH

Research Collaborator, Vessel Trajectory Prediction Pipeline

2026 – Present

Ocean Tracking Network (OTN), Dalhousie University — Prof. Gabriel Spadon, Halifax, NS

- Build and maintain a full AIS data pipeline for vessel trajectory prediction: parser, sliding-window trajectory batching, spatial filtering, an API ingester, an evaluation-metrics module, a visual debugger, and a synthetic data generator.
- Implement sequential motion-prediction stages — a Kalman filter baseline followed by a configurable GRU model with a time-decay gate for irregular AIS ping intervals — informed by a literature review of trajectory-forecasting architectures.
- Collaborate with faculty supervisor Prof. Gabriel Spadon via Git-based version control and code review.

EXPERIENCE

Mobile Application Developer

Jan 2026 – May 2026

Collect (Startup Project), Halifax, NS

- Led a team building a React Native (Expo) platform for local creators, owning technical decisions across Firebase authentication, Cloudflare R2 media storage, and third-party API integrations.
- Shipped a full-featured application including a video-based project feed, real-time event hub, and persistent light/dark UI.

Teaching Assistant

Jan 2024 – Apr 2024

Dalhousie University, Halifax, NS

- Supported first- and second-year students in a required Computer Science course, mentoring on programming fundamentals and problem solving.

PROJECTS

ER Wait-Time Prediction Model

[Nova Scotia Hospitals](#)

- Built and deployed an ML model to predict emergency-room wait times, achieving an 18.5% reduction in wait times across Nova Scotia hospitals.

RAG Hallucination Detection

[50 arXiv Reinforcement Learning Papers](#)

- Built a RAG and faithfulness-evaluation system using FAISS vector search, Hugging Face sentence-transformers, and TinyLlama for generation, with an NLI-based detector flagging unsupported claims and query expansion that measurably improved retrieval recall.

Multi-Agent Delivery Routing

[VRPTW](#)

- Designed a multi-agent system with LLM-based intent extraction converting natural-language delivery requests into structured routing constraints, reaching 100% constraint satisfaction after iterative prompt optimization, validated against 100 bootstrapped synthetic instances.

Explainable ML Audit

[LendingClub Loan Default Prediction](#)

- Audited a LightGBM loan-default model using SHAP, LIME, and partial dependence plots, delivering case narratives and a business-facing slide deck giving non-technical stakeholders a clear, actionable view of model risk.

Student Personal Assistant

[n8n AI Agent](#)

- Built a personal AI agent workflow in n8n that drafts, creates, and sends Gmail messages, tracks a budget, manages Google Calendar events, and answers general and health-related queries end-to-end.

LEADERSHIP & EXTRACURRICULAR

Technical Project Lead

May 2026 – Present

Dalhousie Machine Learning Society, Halifax, NS

- Lead end-to-end delivery of applied ML projects for external clients and research partners, defining technical scope and coordinating cross-functional student teams from data acquisition through deployment.
- Designed and delivered a hands-on workshop, "Build Your First AI Agent," using the smolagents framework, and organized the DMLS Summer Hackathon 2026 across five Atlantic Canada industry problem statements.

Student Representative

Nov 2025 – Present

Dalhousie Machine Learning Society, Halifax, NS

- Serve as liaison between the student community and executive leadership; co-organize workshops, hackathons, and mentorship programs.